

Course code : **CSE3009**
Course title : **No SQL Data Bases**
Module : **3**
Topic : **5**

Redis - Key-Value Database

Objectives

This session will give the knowledge about

- Redis commands on Strings
- Redis commands on Lists

String Commands

Set - Set the string value of a key

```
> set name krishnan
```

OK

Get - Get the value of a key

```
> get name
```

"krishnan"

Mset - Set multiple keys to multiple values

```
> mset name krishnan age 23
```

OK

String Commands

Mget - Get the values of all the given keys

> mget name age

1) "krishnan"

2) "23"

Append – used to append the value along with value available in given key

> append name rao

11

> get name

"krishnanrao"

String Commands

Incr – Increment the INTEGER data type value by 1 of the given key

```
> incr age
```

```
(integer) 24
```

Decr – Decrement the INTEGER data type value by 1 of the given key

```
> decr age
```

```
(integer) 23
```

Incr / Decr will through an error if the value is NOT of an integer type

String Commands

Incrby – Increment the INTEGER data type value by given value of the given key

```
> incrby age 4  
(integer) 27
```

Decrby – Decrement the INTEGER data type value by given value of the given key

```
> decrby age 5  
(integer) 22
```

Incr / Decr by value must be of an integer type

String Commands

Setex – used to set the expiration time in seconds for the key-value pair to be alive

```
> setex login 10 tempuser
```

OK

```
> get login
```

"tempuser"

```
> get login
```

(nil)

String Commands

Setnx – it will insert the key-value pair only if the same key is NOT exist

```
> get name
```

```
"krishnanrao"
```

```
> setnx name newname
```

```
(integer) 0
```

```
> setnx newname radha
```

```
(integer) 1
```

```
> get newname
```

```
"radha"
```


String Commands

Strlen – used to find the length of the value available in the given key

> strlen name

(integer) 11

Del – used to delete a key-value pair

> del newname

(integer) 1

Exists – used to verify whether the key is already exists or not

> exists newname

(integer) 0

String Commands

Rename – used to rename the already exist key. If the key is not available, you will get “ERR no such key” error.

```
> rename name stu_name
```

OK

```
> get stu_name
```

```
"krishnanrao"
```

String Commands

Key pattern – used to find the available key that matches the pattern.

```
> mset ironman 98 spiderman 90 superman 97 batman 95
```

OK

```
> keys s*n    // * multiple any characters
```

1) "spiderman"

2) "superman"

```
> set man 10   // ? Single any character
```

OK

```
> keys m?n
```

1) "man"

String Commands

> mset sam chennai ram delhi tom mumbai

OK

> keys s[ao]m // any one from the list

1) "sam"

> keys r[^o]m //except o anything else

1) "ram"

List Commands

> lpush fruits apple
(integer) 1

//insert apple at front end in fruits list

> lpush fruits orange
(integer) 2

//insert orange at front end in fruits list

> lrange fruits 0 10
1) "orange"
2) "apple"

//display list element from limit 0 to 10

List Commands

> rpush fruits graphs
(integer) 3

//insert graphs at back end in fruits list

> rpush fruits mango
(integer) 4

//insert mango at back end in fruits list

> lrange fruits 0 10

//display list element from limit 0 to 10

- 1) "orange"
- 2) "apple"
- 3) "graphs"
- 4) "mango"

List Commands

> lpop fruits //removes first element and returns from the list
"orange"

> rpop fruits //removes last element and returns from the list
"mango"

> llen fruits //returns the size of the list
(integer) 2

List Commands

> lrange fruits 0 10

1) "apple"

2) "graphs"

> lset fruits 1 banana
OK

//insert banana at index 1 by replacing graphs

> lrange fruits 0 10

1) "apple"

2) "banana"

List Commands

Inserts an element before or after another element in a list

> lrange fruits 0 10

1) "banana"

2) "mango"

3) "apple"

4) "mango"

> lindex fruits 2

"apple"

> linsert fruits after mango orange

5

> lrange fruits 0 10

1) "banana"

2) "mango"

3) "orange"

4) "apple"

5) "mango"

List Commands

Lrem – used to remove the element from the list

> lrange fruits 0 10

- 1) "apple"
- 2) "banana"
- 3) "orange"
- 4) "orange"
- 5) "mango"
- 6) "apple"
- 7) "mango"
- 8) "orange"
- 9) "orange"

> lrem fruits 1 apple
(integer) 1

> lrange fruits 0 10

- 1) "banana"
- 2) "orange"
- 3) "orange"
- 4) "mango"
- 5) "apple"
- 6) "mango"
- 7) "orange"
- 8) "orange"

> lrem fruits -2 orange
(integer) 2

> lrange fruits 0 10

- 1) "banana"
- 2) "orange"
- 3) "orange"
- 4) "mango"
- 5) "apple"
- 6) "mango"

> lrem fruits 0 orange
(integer) 2

> lrange fruits 0 10

- 1) "banana"
- 2) "mango"
- 3) "apple"
- 4) "mango"

List Commands

> lrange fruits 0 10

1) "banana"

2) "mango"

3) "apple"

4) "mango"

> lindex fruits 2 //used to display the element at the specified index
"apple"

Summary

This session will give the knowledge about

- Redis commands on Strings
- Redis commands on Lists
- Reference: <https://redis.io/commands>
- Reference: Redis for Dummies